

Executive Summary: Last week our group decided that the main focus of work needed to be on getting the general planning done for the project. We decided that by the end of the week we would have a list of parts chosen to buy in order to have a general block diagram and possibly schematic next week. This week we did finish finding all of the individual circuits that we needed for the schematic. Those circuits are a GPS module, microphone module, our main microprocessing chip, and our battery for the device. For next week we plan on having a complete block diagram and another in person meeting with our team sponsor to tell him our progress and what we are looking for next week.

Task / Time Take / Completed Y/N

Hardware:

Cody C.

- Progress
 - 2/9 – Meet with the CCEW team to talk about the details of the project (~1 hour)
 - Form Factor: Watch, on the belt, Jewelry
 - Potential Functions: Battery life (Rechargeable), GPS, Button, Gun Shot Detection, Microphone, elevation detection, heart rate sensor (skin to skin contact), and Bluetooth
 - Brained stormed ideas for Hardware features (~1.5 hours)
 - Assigned to research the Gun Shot Detector
 - Find implementations without Arduino Microchip
 - Main functions: Button, Gun Shot Detection, and GPS, if possible
 - Research parts (~3 hours)
 - Microphone and price range

- Found potential microphones to use for the prototype; however, the decibel range found only reached 140 dB. The range for decibels for a gunshot is between 130 dB - 180 dB on the higher end. If there are no products to reach higher than 140 dB, potentially researching into the rise time instead. If there is a big spike/jump, we can potentially emulate the sound of a gunshot. This might have some completions because there are other sounds that can produce a spike in the rise time.

- Discuss with the CCEW team about what decibel range they would like us to reach and detect.

- Button

- Plan
 - Research ideas on how to connect each component
 - Research how to do surface mount soldering
 - Find out where I can solder surface-mount components
 - Create a block diagram for hardware (Rough idea)
 - Create a budget for Hardware parts
 - Find cheaper options and expensive options
 - Weigh in the pros and cons of the products
 - Prep for PDR presentation

Task / Time Take / Completed Y/N

Hardware:

Ryan K.

Progress:

- Last week we met and discussed with Mr. Noonan about the CCEW project and had some specifications clarified such as size, shape, and functionality (1hr). I was tasked with researching a secondary feature of the wearable device, that being GPS integration (3hrs). A module with multiple Global Navigation Satellite Systems (GNSS) and Satellite-Based Augmentation Systems (SBAS) would be best to use as it increases the signal lock time, reliability, and accuracy since the device would be able to connect to multiple different satellites. Some modules offer additional features such as calculating altitude and speed. To improve wearability and portability, modules that require external antennas and USB connections are not recommended but they could still be used for early designs and testing.

Next Week's Tasks:

- Research more GPS modules and narrow down our choices. I will also begin hardware presentation slides for the Preliminary Design Review.

Task / Time Take / Completed Y/N

Software:

Alex H.

Progress:

- **2/9 (1 h 37 min)**
 - Had first client meeting with Mr. Noonan to discuss the requirements of the project. I asked Mr. Noonan the state of the mobile app, and he explained that the clients had a framework for the mobile app, but it was not functional. I also asked Mr. Noonan if we could obtain the source code for the app, but it did not seem like

it would be possible because the app does not function yet. **However, Mr. Nooner says that they are able to alter the app based on what we design for the electronic device, so we do not need to constrain our ideas to exactly tailor with the app.**

- Researched what microcontroller chip to use for the device. I settled on a u-blox NORA-W106-00B chip, which has Wi-Fi and Bluetooth Low-Energy (BLE) capabilities.
- Determined feasibility of using a surface-mounted microcontroller chip on our printed circuit board design to avoid usage of an entire prototype board like Arduino or STM32. In a previous, unrelated, project, I was able to program the u-blox NORA-W106-00B chip without its associated prototype board (Arduino Nano ESP32). I tested if I could still upload software into the chip itself, and I was able to successfully upload code into 10 different NORA chips. **This means that using a surface-mounted microcontroller is feasible.**

- **2/11 (1 h)**

- Provided a USB schematic with NORA-W106-00B chip to the hardware team so they knew how to design the circuit (this circuit is crucial for uploading software into the chip)
- Discussed ideas for detecting gunshots with the team during the weekly meeting. Gunshots range in the 140-170 db range when it comes to noise levels. Most other sounds do not reach this db level, which makes it easier to detect gunshots. To distinguish gunshots and other loud popping noises like firecrackers, a standard

deviation/probability calculation could be implemented to detect the audio signal shape of a gunshot.

- **2/13 (2 h 25 min)**

- Researched more into gunshot detection. The best method would be to use some form of machine learning.
- Discovered that a tiny machine learning (TinyML) model could be installed onto NORA-W106-00B. Processed data that TinyML takes in can be in the form of a C array, which is simple to implement in software.
- Training data can be collected as db levels, which can be processed in some external machine learning software like TensorFlow or Edge Impulse (Edge Impulse is simpler to use) and exported as a C file.

Total Hours Worked: 5 h 2 min

Future Work/Plan:

- Solidify conceptual design of gunshot detection methodology using TinyML model installed on microcontroller.
- Create folders and initial files for two software projects (one project contains scripts for collecting training data, the other project contains the scripts for the actual functionality of the device)
- Create main, hardware, and software version control branches for GitHub repository
- Prepare PDR presentation slides with the rest of the team

Project Manager: This week it was my job to complete team orientated tasks. I scheduled 1 meeting with our team sponsor for the week on Monday. During that meeting I was taking notes

and did most of the point of contact with the sponsor during the teams' meeting. After this I scheduled an interpersonal team meeting for Wednesday where I delegated tasks out and made sure that each member knew their role and goals for the upcoming week. Behind the scenes I was talking with several other teams for the CCEW project to coordinate a meeting in person for next week with our sponsors. This is a work in progress and will continue into next week. I also created several reminders for the team in order to complete all tasks on time for the week.

Jonas W.